Harsh Dhoriyani

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TECHNICAL SKILLS

Languages & Frontend: Java, Python, JavaScript, React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend, Databases & Cloud: Node.js, Express.js, REST API, MySQL, MongoDB, AWS, Docker

Machine Learning & Tools: Scikit-learn, Pandas, NumPy, XGBoost, Git, GitHub, Postman, Jupyter Notebook

PROJECTS

SpectraPI – Spectral Tempering for Associative Memory GitHub

[Python, NumPy, Information Geometry, Optimization, Spectral Analysis]

- Developed an adaptive precision-control framework for a precision-controlled associative memory benchmark, improving retrieval accuracy by **11.9%** through spectral conditioning and covariance-based feature weighting.
- Designed a corruption-aware retrieval pipeline that reconstructed denoised attractor states using confidence-weighted memory retrieval, making the system more robust to noisy and incomplete queries than baseline approaches.
- Applied Hessian-inspired spectral conditioning to reduce anisotropy in high-dimensional memory representations, achieving a **1.29**× reduction in retrieval spread while maintaining stable convergence and retrieval quality.

CivicSpot – Civic Issue Reporting Platform CivicSpot

[React, Node.js, Express.js, MongoDB, Cloudinary, REST APIs]

- Developed a full-stack civic issue reporting platform enabling citizens to report, track, and manage public infrastructure issues through an interactive map interface with real-time status updates.
- Built location-aware issue reporting with geolocation, image uploads, category-based classification, authentication, and role-based dashboards for citizens and administrators, streamlining issue lifecycle management.
- Engineered secure RESTful backend services with JWT authentication, Cloudinary-powered media storage, MongoDB persistence, and responsive React interfaces, delivering seamless issue submission and monitoring workflows.

CircleChat – Real-Time Messaging Platform CircleChat

[React, Node.js, Express, Socket.IO, MongoDB, JWT, Google Gemini AI]

- Built a full-stack real-time messaging engine using **Socket.IO v4** and **MongoDB**, delivering multi-state delivery/read receipts, live typing indicators, and presence tracking across 1-on-1 and group conversations.
- Designed an AI-powered **Catch Me Up** pipeline using Google Gemini to summarize missed messages into recaps and action items, with a heuristic fallback ensuring uptime during API outages.
- Implemented JWT-based session security with version control and single-device login enforcement, alongside rate limiting and MIME-validated file uploads.

EDUCATION

VIT Bhopal University Sept 2023 – Present

B.Tech in Computer Science, Specialization in AI and ML

- GPA: 9.01/10.0
- **Coursework:** Data Structures and Algorithms, Computer Networks, Operating Systems, Database Management Systems, Deep Learning

CERTIFICATIONS

- **Applied Machine Learning in Python** – University of Michigan
- **Data Structures and Algorithms in Java** – Apna College
- **GDG On Campus Solution Challenge** – Google Developer Groups | Hack2skill

ACHIEVEMENTS

- **Runner-up** among **600+ teams** in the national-level Anvil Hackathon 2026 for developing SpectraPI, a geometry-aware associative memory system leveraging precision-controlled retrieval and adaptive spectral conditioning.
- Solved **300+** DSA problems on LeetCode and Codeforces, improving problem-solving and implementation skills.
- Earned an **Elite (Top 1%)** certification with a **93%** score in **Introduction to Internet of Things** – NPTEL, IIT Kharagpur.
- Earned **Google Cloud merchandise** for successful completion of hands-on labs in Google Arcade Program.